

C02-AT1-Scenario Based Programs

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QUESTIONS

Q1
Student Registration System
10 marks**Q2**
E-Commerce Cart System
10 marks**Q3**
Library Management System
10 marks**Q4**
Employee Data Processing
10 marks**Q5**
Online Voting System
10 marks

5/5 answered

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- (a) List three Collection classes applicable for cart management.
(b) Represent how cart items are stored using a List-based structure.
(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.*;
2
3 public class CartSystem {
4
5     public static void main(String[] args) {
6
7         ArrayList<String> cart = new ArrayList<String>();
8
9         cart.add('Laptop');
10        cart.add('Mouse');
11        cart.add('Keyboard');
12
13        Iterator<String> it = cart.iterator();
14
15        System.out.println("Cart Items");
16
17        while(it.hasNext()) {
18            System.out.println(it.next());
19        }
20    }
21 }
```

Input

stdin input

Output

```
Cart Items
Laptop
Mouse
Keyboard
```

Run

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Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
- (b) Design a schema-like representation using Map for storing ISBN and book details.
- (c) Write a Java program using HashMap to:
- Insert book details
 - Retrieve a book
 - Display all books

Your Code

```
1 import java.util.*;
2
3 public class LibrarySystem {
4
5     public static void main(String[] args) {
6
7         HashMap<String,Book> books = new HashMap<String,Book>();
8
9         books.put("101",new Book("Java"));
10        books.put("102",new Book("Python"));
11
12        System.out.println("Book Found:");
13        System.out.println(books.get("101").name);
14
15        System.out.println("\nAll Books");
16
17        for(String key : books.keySet()){
18            System.out.println(key + " " + books.get(key).name);
19        }
20    }
21 }
22
23 class Book{
24
25     String name;
26
27     Book(String name){
28         this.name=name;
29     }
30 }
```

Input

stdin input

Output

```
Book Found:
Java

All Books
101 Java
102 Python
```



Run



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Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```
1 import java.util.*;  
2  
3 public class VotingSystem {  
4  
5     public static void main(String[] args) {  
6  
7         HashSet<Integer> voters = new HashSet<Integer>();  
8         HashMap<String,Integer> votes = new HashMap<String,Integer>();  
9  
10        voters.add(101);  
11        voters.add(102);  
12  
13        votes.put('A',2);  
14        votes.put('B',1);  
15  
16        System.out.println('Results');  
17  
18        for(String key : votes.keySet()){  
19            System.out.println(key + " = " + votes.get(key));  
20        }  
21    }  
22 }
```

Input

Output

```
Results  
A = 2  
B = 1
```



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10 marks

5/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
- Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.*;
2
3 public class EmployeeData {
4
5     public static void main(String[] args) {
6
7         ArrayList<Employee> list = new ArrayList<Employee>();
8
9         list.add(new Employee(101, 'Arun', 60000));
10        list.add(new Employee(102, 'Priya', 45000));
11        list.add(new Employee(103, 'Rahul', 70000));
12
13        Iterator<Employee> it = list.iterator();
14
15        while(it.hasNext()){
16
17            Employee e = it.next();
18
19            if(e.salary > 50000){
20                System.out.println(e.id + " " + e.name + " " + e.salary);
21            }
22        }
23    }
24 }
25
26 class Employee{
27
28     int id;
29     String name;
30     double salary;
31
32     Employee(int id, String name, double salary){
33         this.id = id;
34         this.name = name;
35         this.salary = salary;
36     }
37 }
```

Input

stdin input

Output

```
101 Arun 60000.0
103 Rahul 70000.0
```



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10 marks

5/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection Interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2
3 public class StudentRegistration {
4     public static void main(String[] args) {
5
6         HashSet<Integer> ids = new HashSet<Integer>();
7         HashMap<Integer, Student> map = new HashMap<Integer, Student>();
8
9         Student s1 = new Student(101, 'Arun', 'Java');
10        Student s2 = new Student(102, 'Priya', 'Python');
11
12        ids.add(s1.id);
13        ids.add(s2.id);
14
15        map.put(s1.id, s1);
16        map.put(s2.id, s2);
17
18        Iterator<Student> it = map.values().iterator();
19
20        while(it.hasNext()) {
21            Student s = it.next();
22            System.out.println(s.id + " " + s.name + " " + s.course);
23        }
24    }
25 }
26
27 class Student {
28
29     int id;
30     String name;
31     String course;
32
33     Student(int id, String name, String course) {
34         this.id = id;
35         this.name = name;
36         this.course = course;
37     }
38 }
```

Input

stdin input

Output

```
101 Arun Java
102 Priya Python
```

Run

Save